During my summer internship in 2026, I noticed an interesting phenomenon: the longer I "worked" with my AI agent — or more precisely, my agent paired with a specific LLM — the more clumsy it became. I say "worked" rather than "talked" because I was asking the agent to handle some pretty tricky technical tasks: the agentic CMOS circuit design assigned by the senior engineers, and the human-like GUI operations assigned by our interns. Whenever the agent got stuck in a dead end and started just messing around, I would ask another agent — or even the same agent in a fresh session — for a solution, and I usually got a better answer. When I fed that answer back to my original agent, it seemed as if it had just had an epiphany. This effect is even more pronounced when the paired LLM is a quantized model (ours is FP8) deployed locally on a small GPU.

My experience taught me that as the context gets longer, the agent easily loses focus, but I had no mathematical theory to back up my observation. Then I happened to come across a video (https://b32.tv/zyHDt2X) discussing an article titled "Why your local LLM feels dumber than it is" (https://forum.level1techs.com/t/why-your-local-llm-feels-dumber-than-it-is/253917), and my experience was confirmed. The article shows how subtle, low-level differences in precision (our FP8 vs. unquantized FP16) or in the backend hardware (our small GPU vs. cloud GPUs) trigger a domino effect in tool calling and similar operations, causing the model to execute erroneous commands or exhibit logical drift.

Wait a minute — a subtle difference triggering a domino effect that leads to such a huge difference? That sounds like the butterfly effect, or chaos theory! Out of curiosity, I checked arXiv and did find four papers on related topics:

1. 2503.13530 — Cognitive Activation and Chaotic Dynamics in LLMs
2. 2604.13206 — Numerical Instability and Chaos
3. 2606.02430 — Not All Errors Are Equal
4. 2606.08162 — Silent Failure in LLM Agent Systems

I haven't read the papers in depth, but I now know there really is math behind my experience. What's nice is that when you discuss something with your LLM, you actually get to experience a small butterfly effect of your own.